

Autos

EC3335 Set 11

Chen Xu

Spring 2027

Content

(i) Homeownership

(ii) Rental markets

Housekeeping

PS03 *(due Friday, Nov. 25)*

- Posted now (a couple of minutes ago)

PS04 *(due Wed, Nov. 30)*

- will be posted by Mon next week

TotC book report *(due Thu, Dec. 8)*

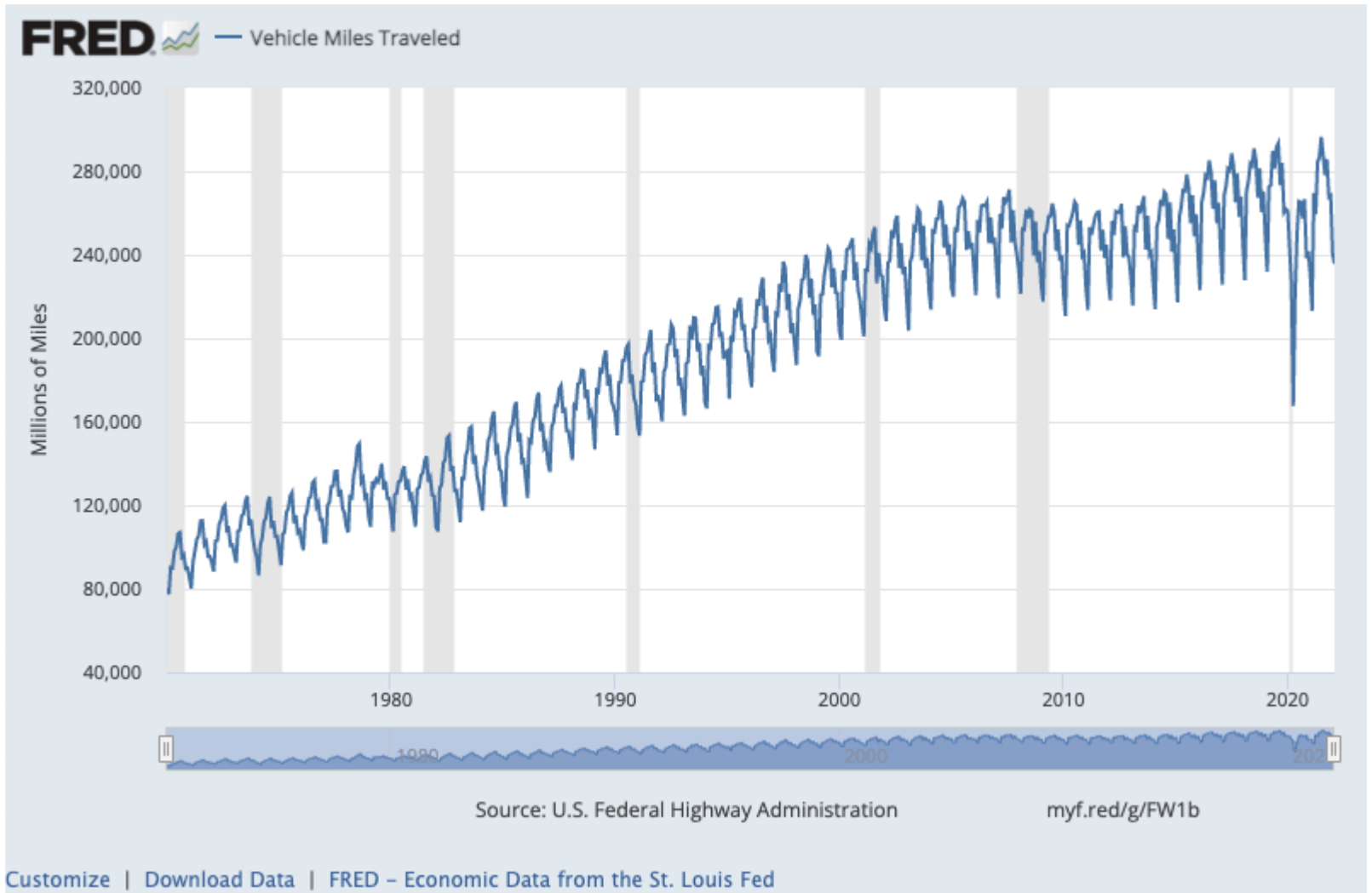
- Short (1000 words) assignment
- rubric posted on canvas and on the course repo
- will be checking for plagiarism

Final *(14:45 Wed, Dec. 7)*

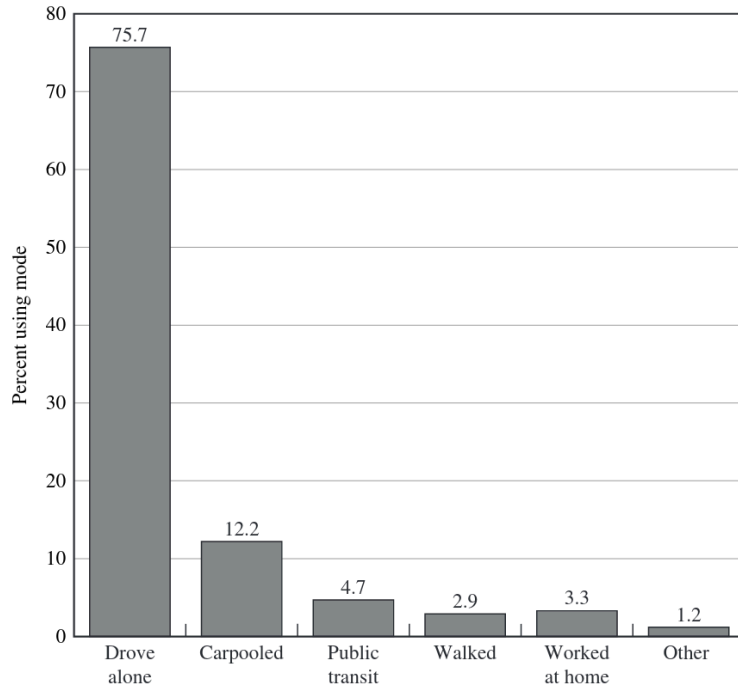
- final schedule
- Comprehensive exam with an emphasis on the newer material

Autos

Auto use in the US



Auto use in the US



Facts about American commuters:

- 88 % use the car
- 5% use public transit
- 2 metro areas have public transit use > 10%
 - NYC
 - Chicago

Climate crisis

A recent [UN scientific report](#) leaves no doubt that humans are responsible for the current climate crisis

Increased frequency of extreme weather including:

- Wildfires
- Winter storms
- Hurricanes
- Heat waves
- Floods
- Droughts

Other consequences include:

- Mass extinctions
- Changes in ocean currents
- Loss of habitat (reefs, rainforest, ice sheets)
- Sea level rise

Global greenhouse gas emissions and warming scenarios

- Each pathway comes with uncertainty, marked by the shading from low to high emissions under each scenario.
- Warming refers to the expected global temperature rise by 2100, relative to pre-industrial temperatures.

Annual global greenhouse gas emissions
in gigatonnes of carbon dioxide-equivalents

150 Gt

100 Gt

50 Gt

Greenhouse gas emissions
up to the present

0

1990 2000 2010 2020 2030 2040 2050 2060 2070 2080 2090 2100

No climate policies

4.1 – 4.8 °C

→ expected emissions in a baseline scenario if countries had not implemented climate reduction policies.

Current policies

2.7 – 3.1 °C

→ emissions with current climate policies in place result in warming of 2.7 to 3.1°C by 2100.

Pledges & targets (2.4 °C)

→ emissions if all countries delivered on reduction pledges result in warming of 2.4°C by 2100.

2°C pathways

1.5°C pathways

Data source: Climate Action Tracker (based on national policies and pledges as of May 2021).
[OurWorldinData.org](https://www.ourworldindata.org) – Research and data to make progress against the world's largest problems.

Last updated: July 2021.
Licensed under CC-BY by the authors Hannah Ritchie & Max Roser.

CO₂ emissions per capita

Carbon dioxide (CO₂) emissions from burning fossil fuels and industrial processes. This includes emissions from transport, electricity generation, and heating, but not land-use change.

Our World
in Data

Table

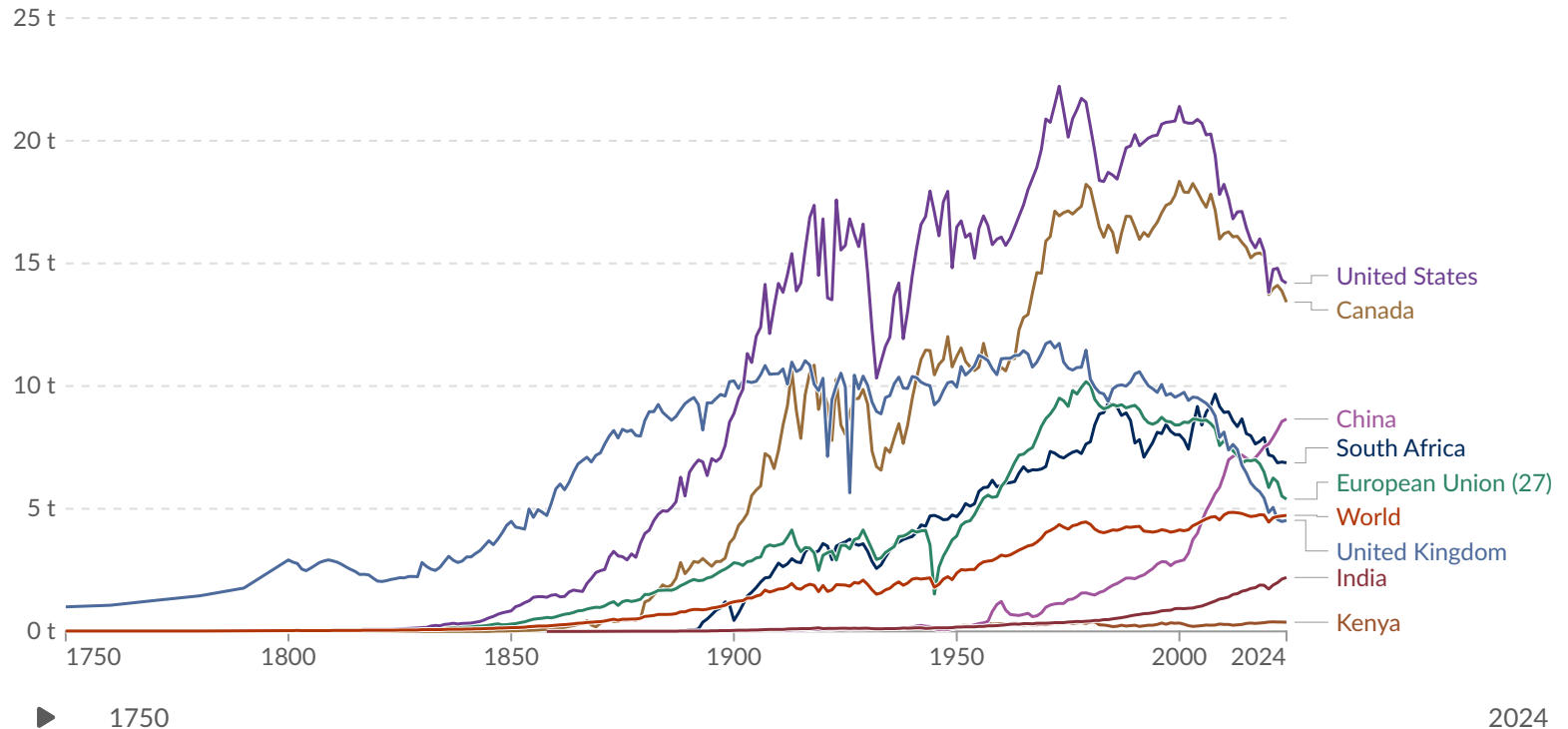
Map

Line

+2

Edit countries and regions

Settings



Data source: Global Carbon Budget (2025); Population based on various sources (2024) –

[Learn more about this data](#)

OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY



Explore the data →

Related: [CO₂ data: sources, methods and FAQs](#) ↗

Annual CO₂ emissions

Carbon dioxide (CO₂) emissions from fossil fuels and industry. Land-use change emissions are not included.

Our World
in Data

Table

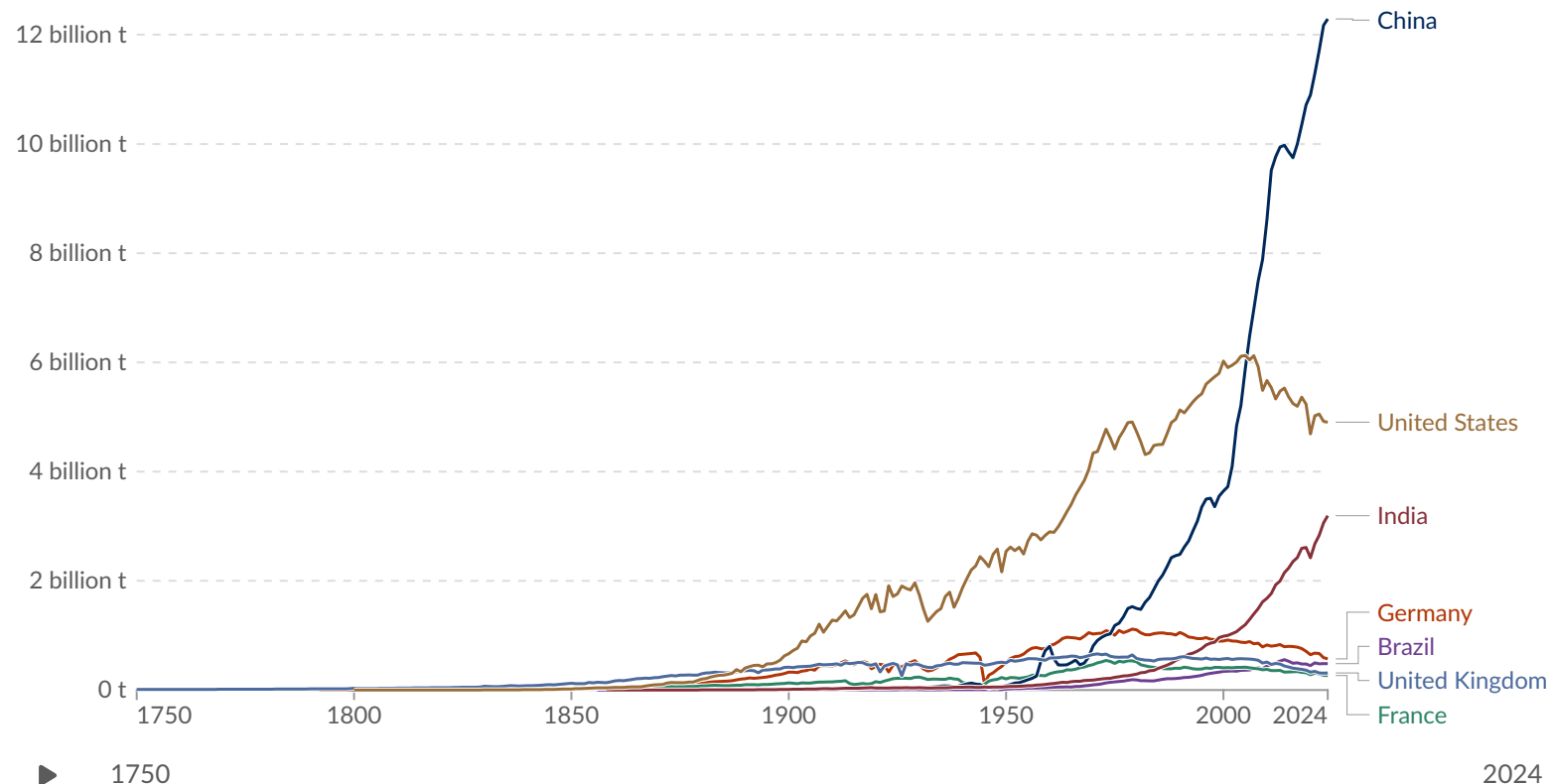
Map

Line

Bar

Edit countries and regions

Settings



Data source: Global Carbon Budget (2025) - [Learn more about this data](#)

OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY



Explore the data →

Related: [CO₂ data: sources, methods and FAQs](#)

Policy questions

CO2 emissions in developing countries are **skyrocketing**

*How do we **reduce** CO2 per capita emissions?*

What axiom is relevant here?

*Do drivers fully **internalize** the costs of CO2 emissions?*

Externalities

Externalities

Recall Axiom 3: Externalities cause inefficiency

What are some externalities from driving?

- Congestion
- Environmental Damage
- Collisions
- Blight (parking lots instead of parks)
- Noise Pollution

How costly is congestion?

Typical commuter spends **47 hours per year** in traffic

- Very high in some metro areas (LA: 93, SF: 72, Atlanta: 67)
- Estimated gasoline cost due to congestion delays: **5 billion per year**
- Time + Gas cost estimate: **63 billion per year**

Externalities

Definition: Marginal Social Cost (MSC)

Added cost to *society* from one extra unit of production/consumption

Note: $MSC \neq MC$

Definition: Marginal Private Cost (MPC)

Added cost to *agent* from one extra unit of production/consumption

MSC is the marginal cost (private) *plus* the marginal *external* cost

Definition: Marginal Social Benefit (MSB)

Added benefit to *society* from one extra unit of production

Definition: Marginal Private Benefit (MPB)

Added benefit to *society* from one extra unit of production

Congestion Externalities

MEC from congestion = $m \cdot v \cdot c$ where

- m is the additional time in traffic from an extra vehicle on the road
- v is the number of other road users
- c is the opportunity cost of time

Note: When there are few cars on the road, m and v are relatively small

- As the number of cars increases, MEC increases (it is nonlinear)
- Also: the above formula makes a strong assumption, what is it?

Congestion pricing

Congestion

How do we fix **Congestion**?

What is the popular answer? **Build more roads**

If you build more

- more space for cars allowing congestion will decrease

What assumption must we make when stating:

"building roads will reduce congestion"

A: The number of drivers will remain the same after the road is built



Congestion

Think of the **incentives** of drivers

(i) People avoid driving because it is costly

- time, gas, maintenance costs

(ii) Building a new road makes it less costly

- time costs reduce

(iii) On the **margin**, people will start to drive when the new road is built

Fundamental law of highway congestion:

When you expand highways, more drivers will substitute toward the new lane

Externalities

Generally when facing **negative externalities**, there exist **two avenues** to reduce private consumption

(ii) Taxes

▮ Increase the cost of consumption

Sometimes referred to as **sin taxes**

Examples:

- tobacco taxes
- alcohol taxes
- luxury taxes

Congestion

If building more roads is not a great solution...

and self-driving cars are not viable...

Pigouvian taxes are an excellent alternative

Main insight:

The social cost of driving exceeds private cost

So how can we raise the MPC such that $\mathbf{MPC = MSC}$

How can we do this? **Tax them.**

- Pigouvian tax specific to reducing road congestion is **congestion pricing**

Externalities: Pigouvian taxes

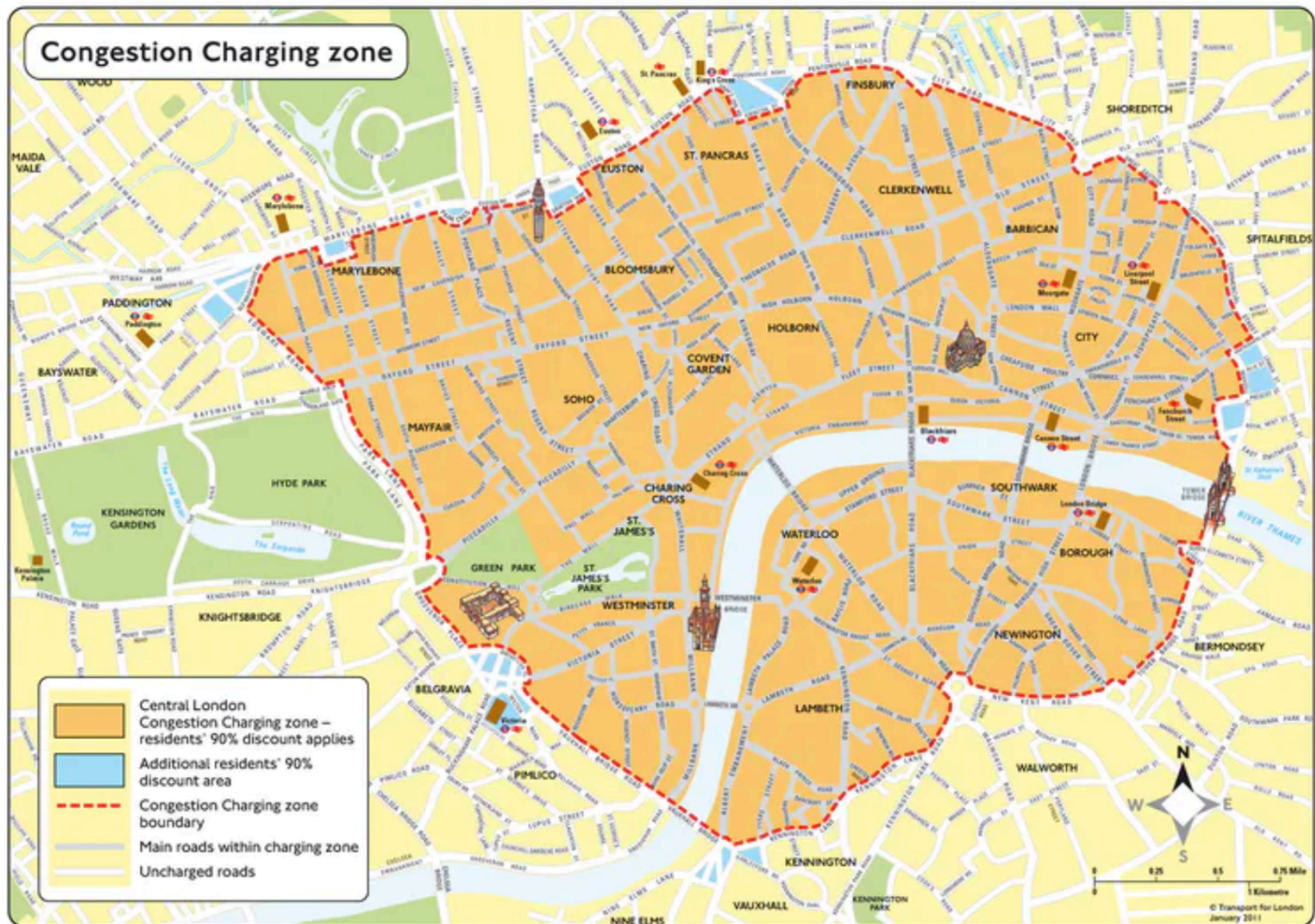
Many cities across the globe have enacted or are planning to enact congestion pricing

- London
- Singapore
- Milan
- Stockholm
- Beijing
- NYC, LA, SF (coming soon)

Results:

Following enactment of congestion pricing in London:

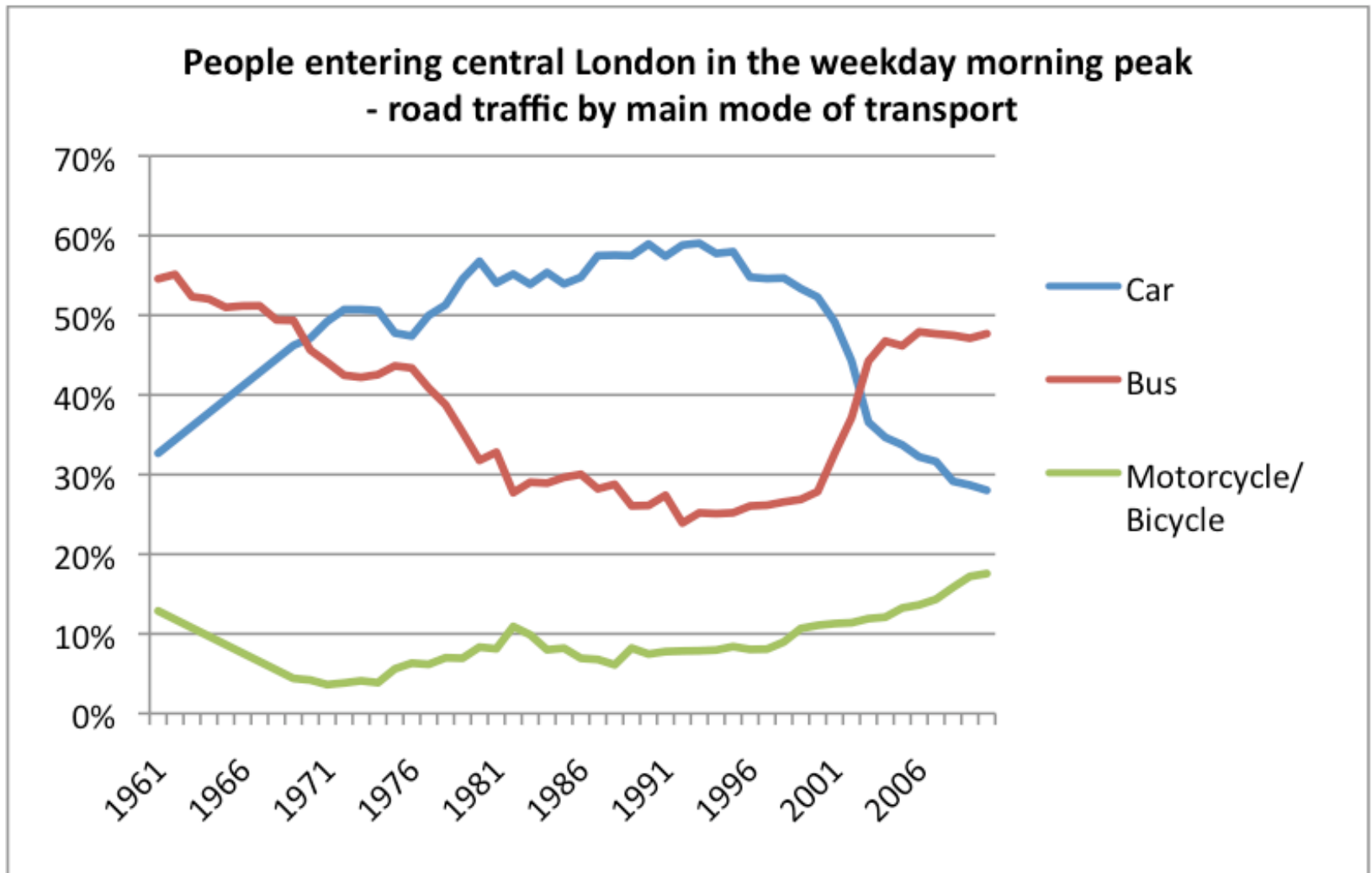
- Traffic was reduced by 15%
- Reduced travel times in the city by 30%



Congestion pricing: London



Congestion pricing: London



Peak vs. Off Period Taxes

Mechanisms

Model demonstrated congestion taxes reduce traffic volume. How?

- (i) Modal substitution: switch to carpool, public transit
- (ii) Switch to off-peak travel
- (ii) Switch route
- (iv) Location decisions: change residence or workplace

Discussion

Congestion taxes sound like a good idea, right? What are the problems?

- Roads aren't always congested. So tax needs to be time-varying. Gets very complicated
- Are all autos charged the same amount (semis and prius?)